

LIGHT RELEASE

A light-emitting diode (LED) emits light when a current is applied. In an LED, a p-type and an n-type semiconductor are placed close together, between electrical contacts. When current flows, excess electrons flow from the n-type to the p-type layer, where they fall into “holes” in the semiconductor and release photons (light). The color of the light emitted depends on the energy required for electrons to cross the band gap (see p.90).

When the positive terminal is connected to the p-type layer and the negative terminal to the n-type layer, an electric current will flow.

**FORWARD
BIAS**

**ENERGY
RELEASE**

When electrons and holes recombine, the electrons—which are at a higher energy level—must release some energy. This is emitted in the form of photons.

